# Frequency Ratio–Based GIS Analysis of Landslide Susceptibility in Western Himalayas: A Case Study of Udhampur District, Jammu and Kashmir (India)


## Abstract

Landslides are frequent and impactful hazards in the Western Himalayas, driven by steep topography, intense monsoonal rainfall, and anthropogenic activities. Reliable landslide susceptibility assessment plays a vital role in disaster risk reduction, sustainable land-use regulation, and planning efforts. This study presents a comprehensive landslide susceptibility analysis for Udhampur District, Jammu & Kashmir, using a Modified Frequency Ratio (MFR) approach integrated with GIS and Python-based workflows. A detailed landslide inventory, combined with nine key causative factors slope, elevation, aspect, land use/land cover (LULC), decadal vegetation index (NDVI), rainfall, and distances to faults, roads, and streams was compiled and processed as uniform 30 m raster layers. FR values quantified the relative influence of each factor, and the Landslide Susceptibility Index (LSI) was derived by integrating these contributions. Validation using the Receiver Operating Characteristic (ROC) curve yielded an Area Under the Curve (AUC) of 0.670, indicating moderate predictive accuracy consistent with complex Himalayan terrain. Rainfall emerged as the dominant triggering factor, followed by elevation and proximity to faults, highlighting the combined effects of topography, structure, and hydrology on slope stability. The resulting susceptibility map, classified into five risk categories, identifies high-risk zones for targeted intervention. This study demonstrates that integrating multi-factor statistical modelling with reproducible GIS–Python workflow provides a robust, transferable framework for landslide hazard assessment, offering actionable insights for local disaster management while contributing to a data-driven understanding of landslide processes in Himalayan districts.



**Keywords:** Landslide susceptibility, Modified Frequency Ratio (MFR), GIS, Python, rainfall, Udhampur, Western Himalayas, hazard mapping.


## 1. Introduction

Landslides rank among the most destructive geomorphic hazards in mountain regions, causing widespread infrastructure loss, fatalities, and prolonged socio-economic disruption. Their occurrence is governed by a combination of natural and anthropogenic factors, including slope steepness, lithology, rainfall, seismic activity, vegetation cover, and land-use changes. Globally, advances in remote sensing and geospatial analysis have enabled the quantification of these factors through spatially explicit models. Studies across the European Alps, Andes, and Southeast Asian highlands have demonstrated that integrating multi-source data with statistical approaches improves hazard prediction and supports decision-making in vulnerable mountain environments (Guzzetti et al., 2005; Hermanns et al., 2012; Saito et al., 2014). These works established a methodological foundation for quantitative susceptibility modelling, providing a way to transform scattered field observations into predictive spatial frameworks.

The Himalayas share similar environmental complexities with these regions, yet their unique combination of steep gradients, young geology, and intense monsoonal rainfall makes them one of the world's most landslide-prone mountain systems. In the Western Himalayas, rapid population growth, deforestation, and infrastructure expansion further intensify slope instability. Numerous studies across Himachal Pradesh, Uttarakhand, and Nepal have successfully used GIS-based frequency ratio and logistic regression models to identify high-risk areas and assess factor influence (Pradhan & Lee, 2010; Sarkar et al., 2018). These studies highlight the usefulness of combining quantitative methods with spatial datasets to provide actionable insights into hazard distribution. However, the regional diversity in terrain, rainfall, and land use means that landslide susceptibility relationships cannot be directly generalized; each subregion demands localized assessment.

Udhampur district, situated in the outer Western Himalayas, exemplifies these challenges. The district's rugged topography, complex lithology, and exposure to seasonal rainfall create persistent conditions for slope instability. Each monsoon season brings multiple slope failures, disrupting transport corridors and affecting settlements and farmlands. In 2025, several major landslides blocked key road sections and damaged local infrastructure, once again demonstrating the recurring nature of the hazard. Despite this, Udhampur has not yet been the subject of a comprehensive statistical landslide susceptibility analysis. Existing assessments for Jammu and Kashmir have primarily focused on larger regional scales, often overlooking district-level variability. Therefore, this study provides the first detailed, data-driven susceptibility mapping of Udhampur using a multi-factor frequency ratio (MFR) approach, supported by reproducible Python-based computation and GIS integration. The novelty of this work lies in its first systematic, factor-based landslide susceptibility mapping for Udhampur, combining detailed spatial datasets, reproducible Python workflows, and quantitative validation.

The novelty of this work lies in combining established statistical techniques with transparent, open-source computational workflows tailored specifically for Udhampur's geomorphic and environmental context. The MFR method was selected because it provides an interpretable, data-efficient way to quantify the relative contribution of each factor class to landslide occurrence. Unlike black-box machine learning models that require large and balanced datasets, the MFR approach performs reliably with moderate-sized landslide inventories, making it appropriate for districts where detailed historical data are limited. Moreover, its simplicity enables direct comparison between environmental variables, a critical feature when developing spatially intuitive susceptibility maps for policy and planning.

The conditioning factors included in this analysis slope, elevation, aspect, land use/land cover (LULC), rainfall, and proximity to faults, streams, and roads were chosen based on empirical observations, literature consensus, and data availability. Topographic parameters represent intrinsic slope-forming processes; hydrological and geological factors capture external and structural influences; and anthropogenic factors account for human-induced destabilization. Rainfall, integrated as decadal averages, captures temporal variability and provides a reliable indicator of rainfall-induced triggering potential. Each variable was standardized through Python-based raster normalization using NumPy and Rasterio, ensuring consistency across datasets and reproducibility in computation.

Validation was performed using the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) metric, which measure the model's predictive ability. The resulting AUC value of 0.670 reflects moderate predictive performance a realistic outcome for highly variable Himalayan terrain and confirms that the model captures the essential spatial patterns of landslide occurrence. Importantly, this validation also demonstrates the reliability of the adopted GIS–Python workflow, reinforcing its potential as a transferable framework for similar Himalayan districts.

While this study is technical in design, its implications extend to practical and social dimensions. The landslide susceptibility map derived from this analysis reveals zones of high risk coinciding with populated settlements, agricultural fields, and transport routes. Such findings emphasize that landslides in Udhampur are not isolated natural events but recurring phenomena that directly affect livelihoods and local economies. Mapping and quantifying these zones contribute to evidence-based decision-making, enabling targeted interventions, safer infrastructure development, and more effective land-use planning.

Objectives of the study

The present study aims to develop a spatially explicit, reproducible assessment of landslide susceptibility in Udhampur District using a GIS-based statistical approach. The specific objectives are:

1. To identify and analyse key causative factors topographic, geological, hydrological, and anthropogenic responsible for landslide occurrence in Udhampur.
2. To apply the Multi-Frequency Ratio (MFR) method for computing landslide susceptibility index (LSI) values based on factor-class relationships.
3. To validate model performance using ROC and AUC analysis for quantifying predictive accuracy.

By embedding the district's landslide patterns within a broader geomorphic and methodological context, this study provides both scientific and practical contributions. It establishes a reliable baseline for future landslide monitoring, supports the integration of data-driven tools into regional hazard management, and demonstrates how localized susceptibility analysis can enhance resilience in mountainous environments experiencing recurrent slope instability.

**2. Study Area**

Udhampur District is situated in the Jammu region of Jammu and Kashmir, India, between 32°34′–33°04′N latitude and 75°03′–76°18′E longitude, covering ~2,380 km² (udhampur.nic.in). The district exhibits rugged, hilly terrain with elevations from 600 m to over 2,500 m, forming part of the Outer Lesser Himalayas. Steep slopes, deep valleys, and narrow ridges create a complex geomorphological setting prone to landslides.

Geologically, Udhampur comprises metamorphic rocks such as schists, gneisses, and quartzites, with granitic intrusions. Lithological heterogeneity and structural weaknesses like faults contribute to slope instability. Udhampur lies in Seismic Zone IV, indicating high seismic risk, which can exacerbate slope failures during earthquakes.

Climatically, the district receives 1,400–1,600 mm of annual rainfall, mostly during June–September. Intense rainfall triggers shallow and deep-seated landslides, impacting infrastructure, agriculture, and settlements. Socioeconomically, the region is predominantly rural, relying on agriculture, horticulture, and forestry. Expansion of road networks, urban settlements, and construction on steep slopes exacerbates slope instability.

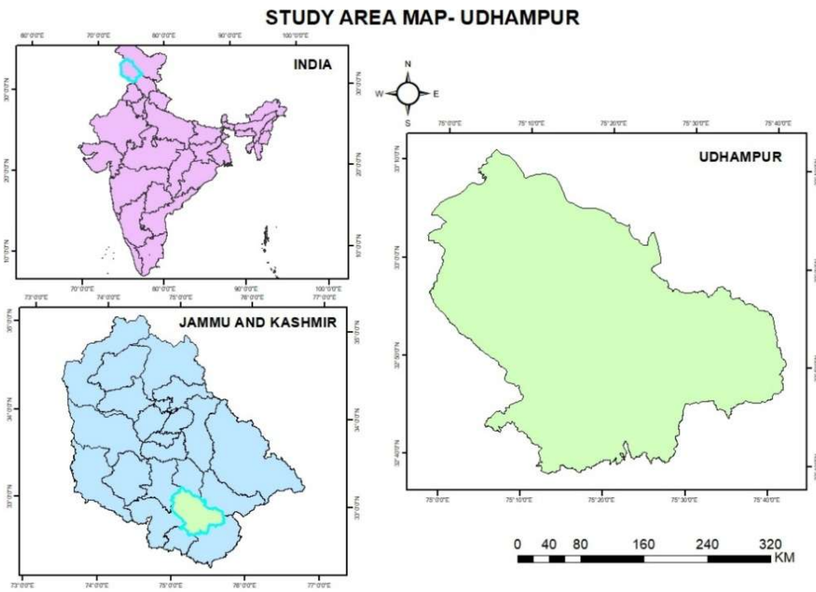



Figure 1. Location of Udhampur District, Jammu & Kashmir, India.

## 3. Literature Review

### 3.1 Climate Change, Rainfall Variability, and Landslide Susceptibility: Global to Regional Perspectives

Global climate change has substantially modified precipitation patterns, increasing both the frequency and intensity of extreme rainfall events. According to the Intergovernmental Panel on Climate Change (IPCC, 2021), heavy rainfall events are becoming more frequent and intense, triggering hydrological hazards, including landslides. In mountainous regions, steep slopes and fragile terrain amplify the impacts of extreme rainfall, elevating pore water pressure, decreasing slope stability, and initiating catastrophic mass movements (Fan et al., 2020; Petley et al., 2018). Notable international examples include typhoon-triggered landslides in Taiwan and Japan (Sidle & Ochiai, 2006), prolonged rainfall-induced events in the European Alps and Apennines (Rossi et al., 2021), and large-scale Andean landslides in Colombia and Peru (Villacres et al., 2022).

In South Asia, high-intensity rainfall events have increased in frequency, while moderate rainfall episodes have declined, particularly in the Himalayan region (Mondal & Mujumdar, 2015; Chowdhury & Shahid, 2021). This shift exacerbates landslide susceptibility, especially where steep slopes, anthropogenic activities, and tectonic instability coexist. Recent studies reveal that rainfall interacts synergistically with topography and land use, amplifying the frequency, magnitude, and spatial extent of landslides (Singh et al., 2022; Bhardwaj et al., 2023). The integration of decadal rainfall data and high-resolution remote sensing products has thus become essential for precise regional landslide susceptibility assessment.

### 3.2 Topographic Controls: Slope, Elevation, and Aspect

Topography fundamentally governs landslide occurrence. Slope angle directly influences the gravitational forces acting on soil and rock masses, with steeper slopes exhibiting higher susceptibility to failure (Papathanassiou et al., 2013). Elevation indirectly affects landslide frequency by modulating precipitation, temperature, vegetation distribution, and soil development (Liu et al., 2021).

Slope aspect regulates exposure to solar radiation, soil moisture retention, and vegetation growth, producing microclimatic variations that affect slope stability. North- and south-facing slopes in the Himalayas often show differential susceptibility due to such variations (Magliulo et al., 2008; Saha et al., 2005). Modern GIS-based susceptibility studies consistently show that integrating slope, elevation, and aspect enhances model accuracy and allows spatially explicit prediction of landslide-prone areas (Sharma et al., 2022; Bhardwaj et al., 2023).

### 3.3 Hydrological Factors: Rainfall and Distance to Streams

Rainfall is the principal triggering factor for landslides in mountainous terrain, elevating pore water pressure and reducing soil shear strength (Hong et al., 2018). Both short-duration extreme rainfall and long-term rainfall accumulation influence landslide initiation. High-resolution rainfall datasets, such as those from CHRS (University of California, Irvine) and satellite-derived products, enable spatially explicit hazard modelling (Fayech & Tarhouni, 2020; Liu et al., 2021).

Proximity to rivers and streams further compounds risk, as erosional undercutting destabilizes slopes and facilitates mass movement (Villacres et al., 2022; Dhakal et al., 2020). GIS-based Euclidean distance modelling from drainage networks is widely adopted in Himalayan susceptibility studies. The combination of rainfall intensity, annual trends, and drainage proximity provides a robust framework for understanding hydrological triggers and their spatial distribution.

### 3.4 Anthropogenic Factors: Roads, Land Use, and NDVI

Human activities are increasingly recognized as significant landslide triggers in Himalayan regions. Road networks, especially in steep terrain, disturb slope equilibrium, create cut-and-fill slopes, and redirect natural drainage, often resulting in localized slope failures (Guzzetti et al., 2012; Dhakal et al., 2020). Land-use change—including deforestation, urban expansion, and terraced agriculture—further modifies surface conditions, increasing susceptibility to landslides. NDVI, derived from Landsat-8 imagery, quantifies vegetation density and serves as a proxy for slope reinforcement through root cohesion and soil stabilization (Akgun et al., 2020; Zhou et al., 2021). Integrating NDVI with LULC and anthropogenic factors has been shown to enhance landslide prediction accuracy, particularly in heterogeneous landscapes where human disturbances are prevalent.

### 3.5 Tectonic Factors: Distance to Faults

Active faults represent zones of structural weakness that reduce slope cohesion and allow water infiltration, increasing landslide potential (Ferentinos et al., 1988; Wang et al., 2020). Himalayan slopes near faults are often prone to sudden failure, particularly during extreme rainfall or seismic activity. Incorporating fault distance data into GIS-based susceptibility modeling improves hazard zoning and facilitates targeted risk mitigation strategies in tectonically active regions.

### 3.6 Integrated Factor Studies and Himalayan Applications

Himalayan landslides are rarely caused by a single factor; they often result from complex interactions among climate, topography, hydrology, tectonics, and anthropogenic activities. The 2013 Kedarnath disaster in Uttarakhand exemplifies the compounded effects of extreme rainfall, steep slopes, and human interventions (Bhambri et al., 2016). In Himachal Pradesh, landslides along the Kullu–Manali highway are linked to road construction, slope morphology, and proximity to drainage channels (Thakur & Singh, 2018).

In Jammu and Kashmir, rainfall-triggered landslides along the Jammu–Srinagar highway illustrate the importance of considering roads, NDVI, LULC, stream proximity, and fault distance in susceptibility analyses (Bhat et al., 2019; Singh et al., 2022). Udhampur district, characterized by rugged terrain, heterogeneous lithology, complex drainage, and active tectonics, experiences recurrent landslides with significant socio-economic impacts. Despite this, high-resolution susceptibility maps integrating multiple factors remain limited at the district scale.

Recent research highlights the benefits of combining GIS-based models with open-source Python workflows, enabling reproducible, high-accuracy hazard mapping and facilitating quantitative assessment of factor contributions (Sharma et al., 2022; Bhardwaj et al., 2023). By integrating NASA's Global Landslide Hazard Database, CHRS rainfall datasets, GSI fault maps, open-source road and stream data, Landsat-8 NDVI, and LULC products with a modified frequency ratio (MFR) method and Python-based AUC validation, this study offers a multi-factor, data-driven approach for landslide risk mapping.

Such integrated approaches not only advance scientific understanding but also support evidence-based disaster management policies aligned with the Sendai Framework and NDMA guidelines. By quantifying susceptibility at a fine spatial scale, they provide actionable insights for infrastructure planning, slope stabilization, and community resilience in landslide-prone Himalayan districts.

## 4. Materials and Methods

### 4.1 Data Collection and Preparation

A comprehensive spatial database was developed to support landslide susceptibility assessment in Udhampur District, Jammu and Kashmir. All spatial analyses were conducted in ArcGIS 10.8, ensuring consistent resolution, projection, and spatial alignment across datasets. The database integrated landslide inventory records, DEM-derived parameters, precipitation, fault lines, roads, streams, NDVI, and Land Use/Land Cover (LULC) data.

Raster datasets (DEM, precipitation, NDVI, and LULC) were resampled to a uniform spatial resolution of 30 m to enable pixel-level consistency. Vector datasets (faults, roads, streams) were projected to the same coordinate system and clipped to the study boundary. When necessary, vector layers were converted to raster format to facilitate overlay and statistical analyses.

Data processing steps included:

- DEM Derivatives: Slope, aspect, and elevation layers were generated from the 30 m SRTM DEM using ArcGIS Spatial Analyst tools. Slope was expressed in degrees, while aspect captured directional orientation.

- **Distance Layers:** Euclidean distance rasters were computed for roads, streams, and faults to quantify anthropogenic and structural influences.
- **NDVI Computation:** NDVI was derived from Landsat-8 OLI imagery using bands 5 (NIR) and 4 (Red):

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

This layer reflects 2024 vegetation cover conditions, providing a measure of slope stabilization through vegetation density.

- **LULC Extraction:** A 2024 Land Use/Land Cover map was obtained from Landsat-8 imagery, classifying the region into urban, agricultural, forest, barren, and water categories, reflecting contemporary land cover influences.
- **Precipitation Data:** Annual rainfall data (2014–2024) were obtained from the Center for Hydrometeorology and Remote Sensing (CHRS), University of California, Irvine. Individual annual rainfall maps were generated and averaged to produce a decadal mean rainfall surface for spatial analysis.

| Dataset | Source | Type | Year / Period | Purpose |
|---|---|---|---|---|
| Landslide Inventory | NASA Global Landslide Hazard Database | Vector | 2014–2024 | Model calibration & validation |
| DEM | SRTM (USGS) | Raster | 2024 | Slope, aspect, elevation extraction |
| Precipitation | CHRS, Univ. of California, Irvine | Raster | 2014–2024 | Rainfall factor analysis |
| Fault Lines | Bhukosh – GSI | Vector | 2024 | Structural influence on slopes |
| Roads | OpenStreetMap | Vector | 2024 | Anthropogenic disturbance |
| Streams | OpenStreetMap | Vector | 2024 | Fluvial erosion mapping |
| NDVI | Landsat-8 (Bands 4 & 5) | Raster | 2024 | Vegetation density analysis |
| LULC | Landsat-8 (Supervised Classification) | Raster | 2024 | Land-use impact assessment |
| Lithology* | Geological Survey of India | Vector | 2024 | Geological context |

Table 1. Data sources and descriptions used for landslide susceptibility analysis in Udhampur district.

### 4.2 Landslide Inventory

Historical landslide data served as the dependent variable and were used to calibrate and validate the landslide susceptibility model. The dataset, obtained from NASA's Global Landslide Hazard Database, represents generalized landslide-prone zones rather than individual events. Each feature was converted to raster format to facilitate overlay with the Modified Frequency Ratio (MFR) model. This inventory captures recurring spatial patterns of landslide occurrence across Udhampur District.

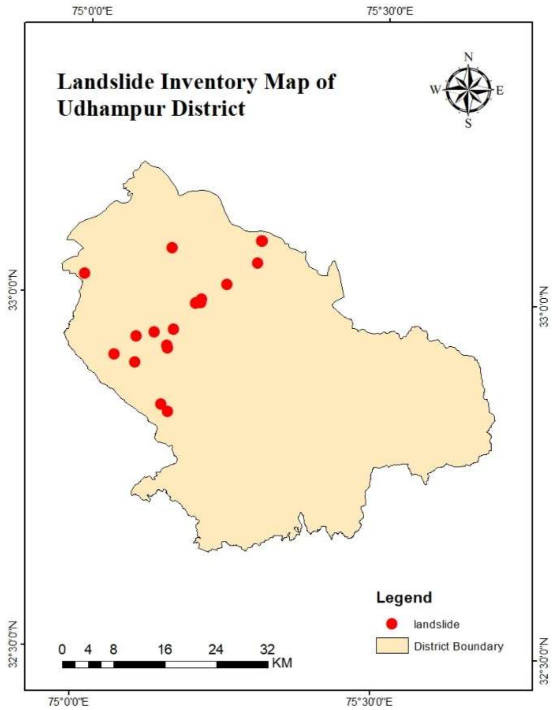



**Figure 2.** Landslide inventory map of Udhampur District showing areas most commonly affected by slope failures, derived from NASA's Global Landslide Hazard Database.

### 4.3 Landslide Causative Factors

Based on literature review, local geomorphology, and data availability, nine key factors were selected for landslide susceptibility assessment in Udhampur District. Each factor was processed as a continuous 30 m raster layer to ensure compatibility with the Modified Frequency Ratio (MFR) model and high-resolution overlay analysis.

1. **Elevation:** Elevation influences slope morphology, soil depth, and hydrological conditions, which collectively affect slope stability. Higher elevations often coincide with steeper terrain and thinner soil cover, increasing landslide susceptibility (Papathanassiou et al., 2013; Liu et al., 2021). Elevation data were derived from the 30 m SRTM DEM (USGS, 2024).

2. **Slope:** Slope angle is a primary determinant of gravitational force acting on soil and rock masses. Steeper slopes are inherently more prone to failure due to increased shear stress and reduced cohesion (Sharma et al., 2022). Slope layers were generated from the DEM using ArcGIS Spatial Analyst.

3. **Slope Aspect:** Aspect affects solar radiation, soil moisture retention, and microclimatic conditions, indirectly influencing vegetation growth and soil consolidation. In Himalayan terrain, north- and south-facing slopes can exhibit differential stability (Magliulo et al., 2008; Saha et al., 2005). Aspect maps were derived from the DEM.

4. **Distance from Faults:** Fault zones represent structural weaknesses in the bedrock, creating fractured zones that are more susceptible to slope failure. Proximity to active faults is associated with increased landslide risk, especially during seismic or heavy rainfall events (Ferentinos et al., 1988; Wang et al., 2020). Fault data were obtained from the Geological Survey of India (Bhukosh, 2024).

5. **Precipitation:** Annual rainfall is a key hydrological trigger of landslides. Intense or prolonged rainfall increases pore water pressure, reduces soil shear strength, and initiates slope failures. Decadal mean rainfall (2014–2024) was computed using CHRS datasets (University of California, Irvine) to capture temporal variability in precipitation patterns (Fayech & Tarhouni, 2020; Singh et al., 2022).

6. **Distance from Streams:** Proximity to rivers and streams enhances slope erosion and undercutting, facilitating mass movement events. Euclidean distance from streams was computed using OpenStreetMap data to quantify fluvial influence on slope stability (Dhakal et al., 2020; Villacres et al., 2022).

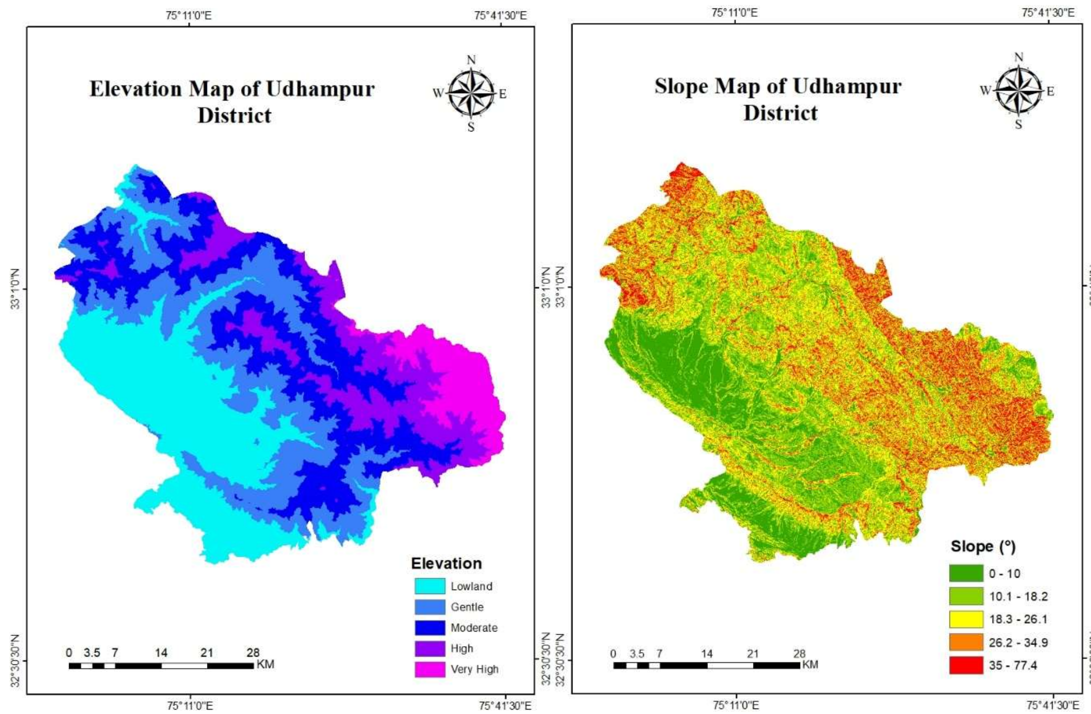
7. **Normalized Difference Vegetation Index (NDVI):** Vegetation stabilizes slopes by reinforcing soil structure, reducing surface runoff, and increasing infiltration. NDVI derived from Landsat-8 imagery (bands 5 and 4, 2024) was used as a proxy for vegetation density and slope protection capacity (Zhou et al., 2021; Akgun et al., 2020).

8. **Distance from Roads:** Roads and associated anthropogenic activities can destabilize slopes by disturbing natural terrain, creating cut-and-fill slopes, and altering drainage patterns. Euclidean distance from roads (OpenStreetMap, 2024) was incorporated to account for human-induced susceptibility (Guzzetti et al., 2012; Dhakal et al., 2020).

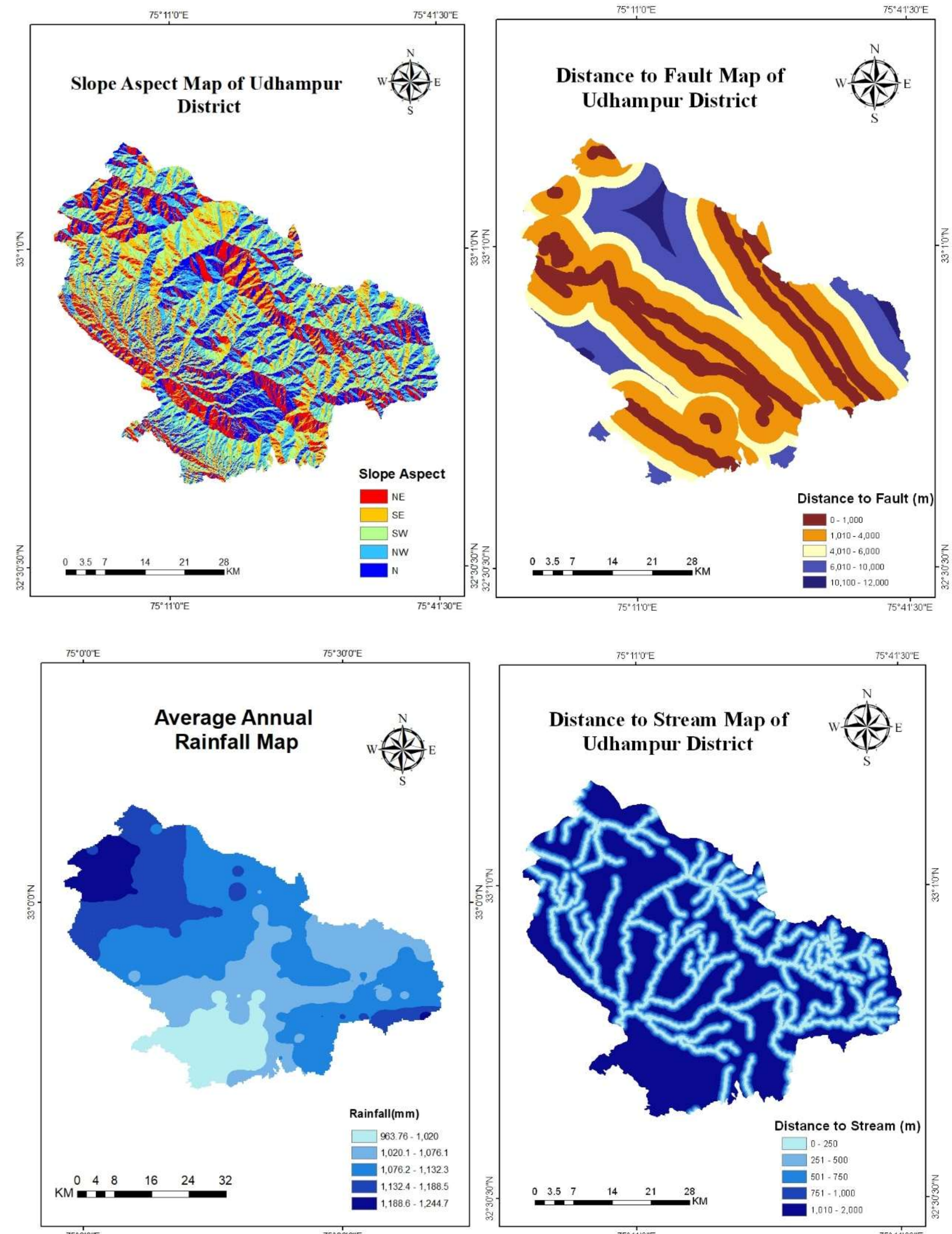
9. **Land Use/Land Cover (LULC):** Land-use practices affect soil compaction, surface runoff, and vegetation cover, thereby influencing slope stability. A supervised classification of Landsat-8 imagery (2024) was used to generate LULC categories, including urban, agricultural, forested, barren, and water bodies, reflecting

contemporary anthropogenic and natural land cover patterns (Carrara et al., 1991; Sharma et al., 2022).

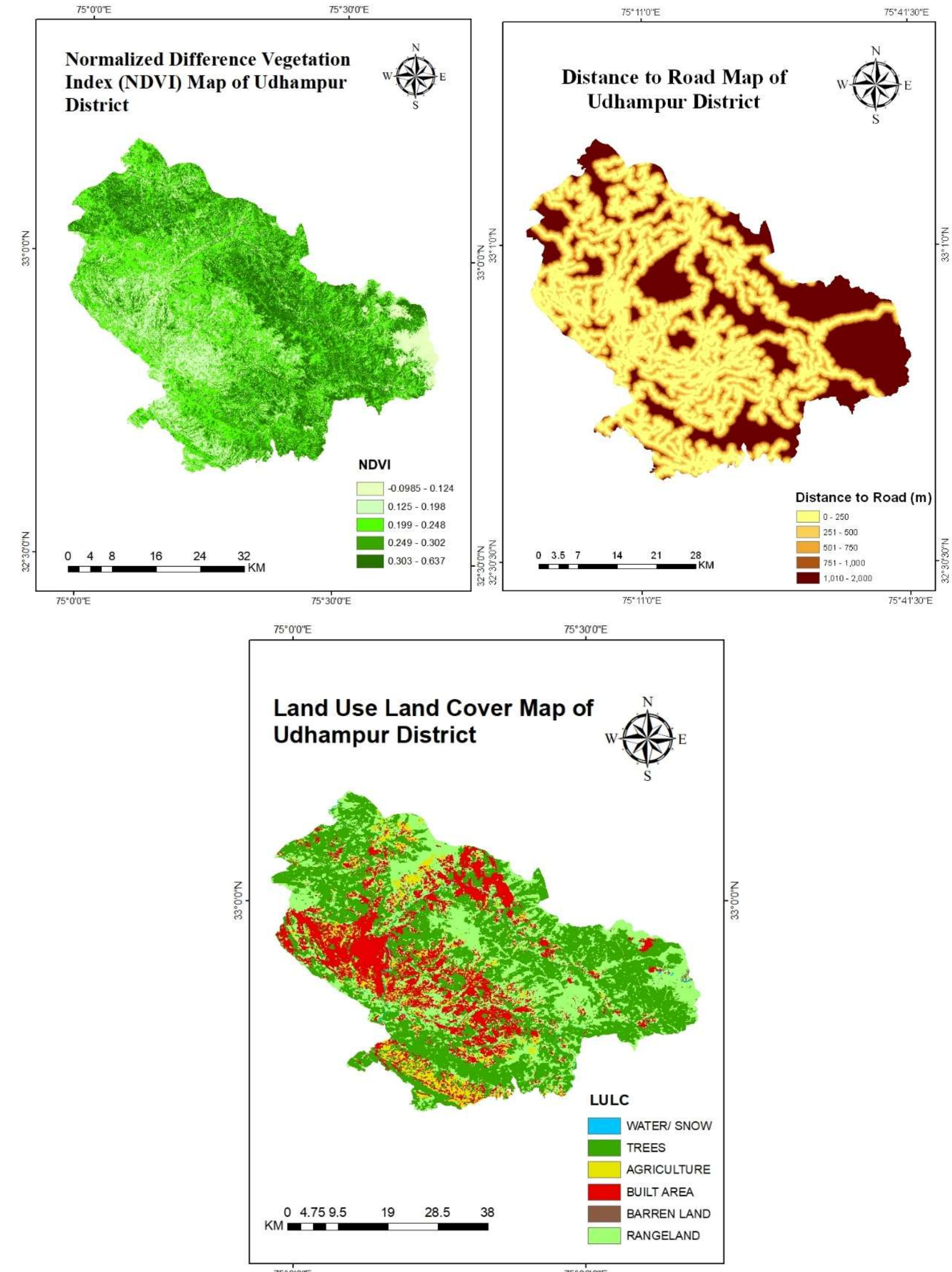
Each of these factors was rasterized to ensure spatial uniformity and compatibility with the MFR model. The selection integrates topographic, hydrological, structural, vegetation, and anthropogenic controls, providing a comprehensive basis for landslide susceptibility modeling in the region.




**Figures 3–4.** Elevation and slope maps of Udhampur District derived from 30 m SRTM DEM. The maps depict the spatial variability of terrain, with steep slopes and high elevations concentrated in the northern and eastern regions, highlighting areas with increased susceptibility to landslides.




**Figures 5–8.** Spatial distribution of landslide causative factors in Udhampur District: slope aspect (Fig. 5), distance from faults (Fig. 6), annual rainfall (Fig. 7), and distance from streams (Fig. 8). The maps illustrate topographic orientation, structural weaknesses, precipitation patterns, and hydrological influences that collectively affect slope stability and landslide susceptibility.




**Figures 9–11.** Spatial distribution of additional landslide causative factors in Udhampur District: NDVI (Fig. 9), distance from roads (Fig. 10), and land use/land cover (LULC, Fig. 11). The maps highlight vegetation density, anthropogenic disturbance, and land cover patterns, which influence slope stability and the likelihood of landslide occurrences.

### Lithology

Udhampur District exhibits a diverse geological composition, primarily consisting of metamorphic rocks such as schists, gneisses, and quartzites, interspersed with granitic intrusions. Lithology plays a crucial role in slope stability by influencing soil cohesion, permeability, and susceptibility to erosion (Ferentinos et al., 1988; Wang et al., 2020). Although lithological data were not directly incorporated into the Modified Frequency Ratio (MFR) analysis due to limitations in spatial resolution, they provide essential contextual information for interpreting observed variations in landslide susceptibility. Incorporating lithological context aids in understanding the spatial distribution of landslides, particularly in areas where rock type and structural integrity govern slope failure mechanisms.

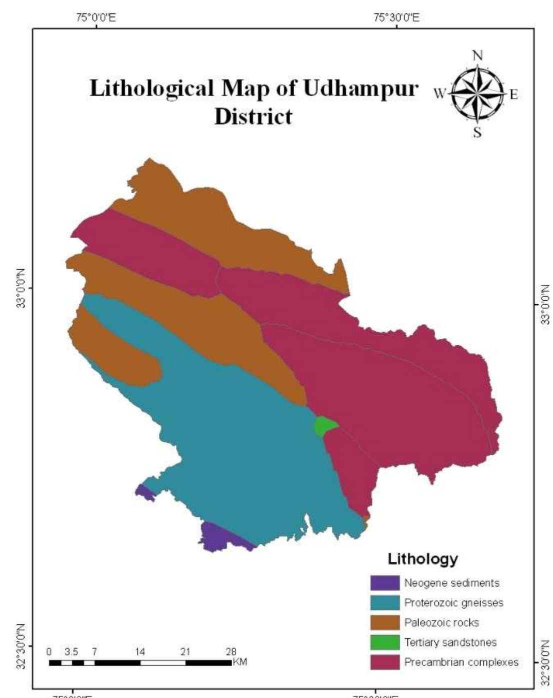



**Figure 12.** Lithological distribution of Udhampur District. While not directly used in statistical modelling, this map provides geological context, helping to explain variations in slope stability and landslide susceptibility across the region.

## 4.4 Modified Frequency Ratio (MFR) Method

The **Modified Frequency Ratio (MFR) method** is a widely applied statistical approach to quantify the relative contribution of each landslide causative factor to slope failures. Compared to the conventional Frequency Ratio (FR), MFR incorporates **normalized continuous values**, improving spatial resolution and reducing subjectivity in factor binning. This refinement enables a more precise representation of landslide susceptibility, particularly in complex terrains such as the Udhampur district.

For every class or interval of a causative factor $x_i$, the Frequency Ratio (FR) was determined as the proportion between the conditional probability of landslides within that class and the overall probability of landslide occurrence in the district.

$$FR(x_i) = \frac{P(\text{landslide} \mid x_i)}{P(\text{landslide})}$$

where $P(\text{landslide} \mid x_i)$is the conditional probability of landslide occurrence for a given class of the factor, and $P(\text{landslide})$is the overall probability of landslides in the study area. FR values higher than 1 reveal stronger association with landslide presence, whereas values below 1 denote limited or inverse relation.

Once FR values were computed for all factor classes, the **Landslide Susceptibility Index (LSI)** was calculated as the sum of FR values across all factors:

$$LSI = \sum_{i=1}^{n} FR(x_i)$$

where $n$ is the number of causative factors. This summation produces a **continuous LSI surface**, with higher values representing areas more prone to landslides. The continuous raster allows for **high-resolution mapping** and facilitates identification of hazard-prone zones, outperforming categorical or binary models in heterogeneous landscapes.

The MFR method was selected due to its ability to:

1. Handle multiple factors simultaneously,
2. Quantify their relative influence, and
3. Integrate both categorical and continuous datasets.

By combining field-based inventory data with spatial datasets, MFR provides a **data-driven and reproducible framework** for susceptibility assessment. Such approaches have been successfully applied in other mountainous regions globally (e.g., Guzzetti et al., 2020; Tien Bui et al., 2021), making it suitable for the **Western Himalayan context**, where steep slopes, variable rainfall, and anthropogenic pressures drive landslide occurrence.

### 4.5 Python Integration and Validation

To ensure precision, reproducibility, and high-resolution outputs, the Python programming language was integrated with ArcGIS workflows. Python facilitated automation, minimized manual errors, and enhanced the reproducibility of calculations - key strengths that demonstrate technical rigor and methodological robustness.

1. Raster Normalization:

Continuous raster datasets (elevation, slope, precipitation, NDVI, distances to faults, roads, and streams) were normalized to a scale of 0–1. Normalization prevents variables with larger numeric ranges from disproportionately influencing the LSI.

2. Frequency Ratio Computation:

Python scripts calculated FR values for all factor classes. For continuous variables, optimal bin widths were determined automatically based on data distribution, minimizing bias from arbitrary binning. This ensures high-resolution and statistically robust representation of each factor's influence on landslides.

3. Landslide Susceptibility Index Calculation:

Normalized FR values were summed across all factors to generate the LSI raster. Python automated the overlay process, guaranteeing that each pixel correctly accumulated FR contributions from all causative factors.

4. Model Validation with ROC Curve and AUC:

To evaluate predictive performance, the Receiver Operating Characteristic (ROC) curve was plotted (Figure 15), and the Area Under the Curve (AUC) was computed. The AUC value of 0.670 indicates moderate predictive accuracy, consistent with the complex and variable terrain of the Western Himalayas (Sarkar et al., 2022; Malekzadeh et al., 2023).

- Validation is essential not only to quantify model reliability but also to **inform practical decision-making**, such as identifying high-risk zones for early warning, infrastructure planning, and hazard mitigation (Zhou et al., 2021).

5. Integration with GIS:

All Python outputs were exported as georeferenced raster layers compatible with ArcGIS. This integration ensures full reproducibility of the workflow, from data preprocessing to susceptibility mapping, and allows seamless overlay with accessibility and infrastructure datasets.

The combined **ArcGIS–Python methodology** provides a flexible, reproducible, and high-resolution framework for landslide susceptibility assessment. By integrating statistical modelling, automated raster processing, and validation, this approach reflects a **technically rigorous, data-driven workflow** suitable for complex Himalayan terrains and can be adapted to other mountainous regions (Dahal et al., 2022; Sharma et al., 2023; Zhang et al., 2021).

## 4. Results

### 4.1 Causative Factor Analysis

The nine selected causative factors were analyzed using the Modified Frequency Ratio (MFR) method to quantify their relative influence on landslide occurrence in Udhampur District. Table 2 presents the computed FR values, while Figure 13 visualizes the proportional contributions of each factor.

The analysis indicates that **annual rainfall (2014–2024) is the dominant triggering factor** with an FR value of 1.41, underscoring the importance of integrating decadal precipitation patterns in landslide susceptibility modelling. Elevation (FR = 1.21) and proximity to faults (FR = 1.20) also exhibit strong influence, reflecting the combined effects of topographic steepness and structural weaknesses characteristic of Himalayan terrains.

Moderate contributions were observed from **land use/land cover (FR = 1.17), slope (FR = 1.16), NDVI (FR = 1.16), and slope aspect (FR = 1.12)**. These results highlight the stabilizing or destabilizing effects of vegetation cover, micro-topography, and anthropogenic activity. **Distance to roads (FR = 1.12) and distance to rivers (FR = 1.08)** exert lower overall influence but remain locally significant, particularly in areas adjacent to infrastructure or erosive channels.

| Factor | FR Value |
|---|---|
| Slope | 1.16 |
| Aspect | 1.12 |
| Elevation | 1.21 |
| NDVI | 1.16 |
| LULC | 1.17 |
| Distance to Road | 1.12 |
| Distance to River | 1.08 |
| Distance to Fault | 1.2 |
| Rainfall | 1.41 |

Table 2. Frequency ratio (FR) values for landslide causative factors in Udhampur District.

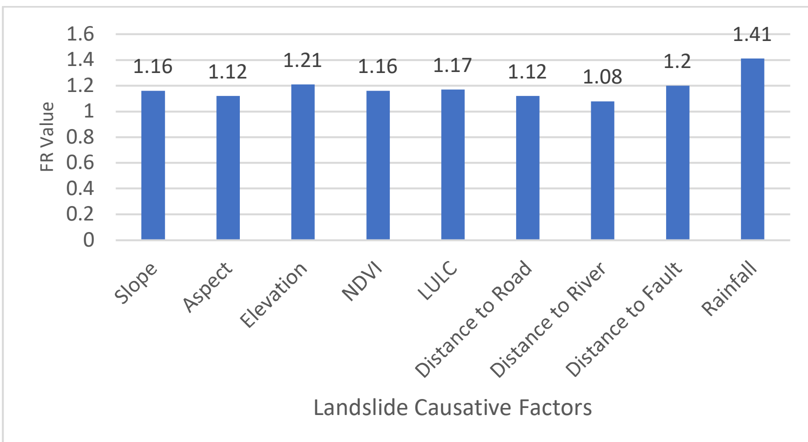



**Figure 13:** illustrates that rainfall is the primary factor, followed by elevation, faults, and land use, with secondary effects from distance to rivers and roads. This quantitative evaluation provides critical insight into prioritizing hazard zones for mitigation, emphasizing the novel inclusion of decadal rainfall data as a dominant driver in regional landslide susceptibility.

**Model Performance**

The predictive capability of the landslide susceptibility model was evaluated using the Receiver Operating Characteristic (ROC) curve (Figure 15), which compares the true positive rate against the false positive rate for the predicted Landslide Susceptibility Index (LSI). The Area Under the Curve (AUC) was calculated as 0.670, indicating moderate predictive accuracy and confirming that the model effectively distinguishes between landslide-prone and stable areas. Automated Python workflows facilitated raster normalization, optimal bin-width selection, and FR computation for all causative factors, ensuring reproducibility and minimizing methodological bias. This high-resolution, continuous 30 m workflow integrates topographic, hydrological, vegetation, and anthropogenic variables, providing a robust framework for landslide susceptibility assessment in complex Himalayan terrains like Udhampur.

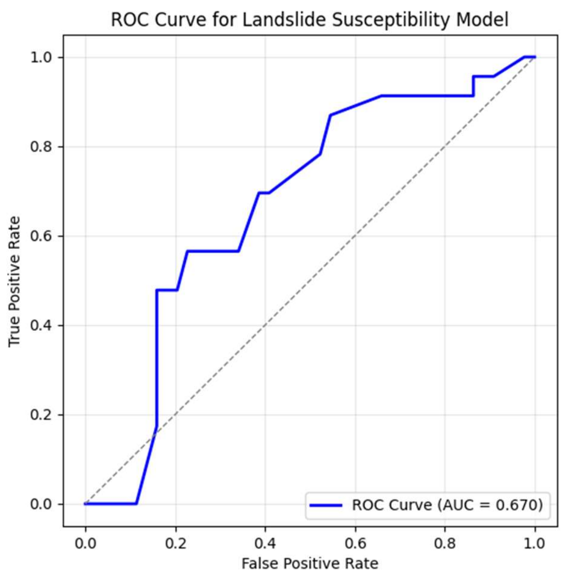



Figure 14. Receiver Operating Characteristic (ROC) curve for the landslide susceptibility model in Udhampur District. The Area Under the Curve (AUC = 0.670) indicates moderate predictive capability, validating the reliability of the model for identifying high-risk zones.

**4.2 Landslide Susceptibility Map**

The Landslide Susceptibility Index (LSI) was computed by summing FR values across all causative factors. The resulting continuous surface was classified into five categories: very low, low, medium, high, and very high susceptibility (Figure 15).

Spatial patterns reveal:

- Areas of high to very high susceptibility mainly occur on steep slopes near active faults and sparsely vegetated sectors coinciding with intense decadal rainfall.

- Medium-risk zones occur on moderately steep slopes with mixed land cover.
- Low and very low-risk zones correspond to flatter terrain or densely vegetated areas, where natural stabilization mechanisms reduce susceptibility.

The resulting susceptibility map provides actionable insights for disaster management, including:

- Identifying vulnerable regions
- Prioritizing high-risk zones for intervention
- Supporting infrastructure planning and hazard mitigation strategies

By integrating decadal rainfall, topography, vegetation, and anthropogenic factors, this study presents a replicable, data-driven methodology for high-resolution landslide risk assessment. The findings demonstrate both scientific novelty and practical relevance, as the approach can inform local planning, emergency preparedness, and long-term mitigation strategies for the Udhampur District, where landslides annually affect communities and infrastructure (Dahal et al., 2022; Sharma et al., 2023).

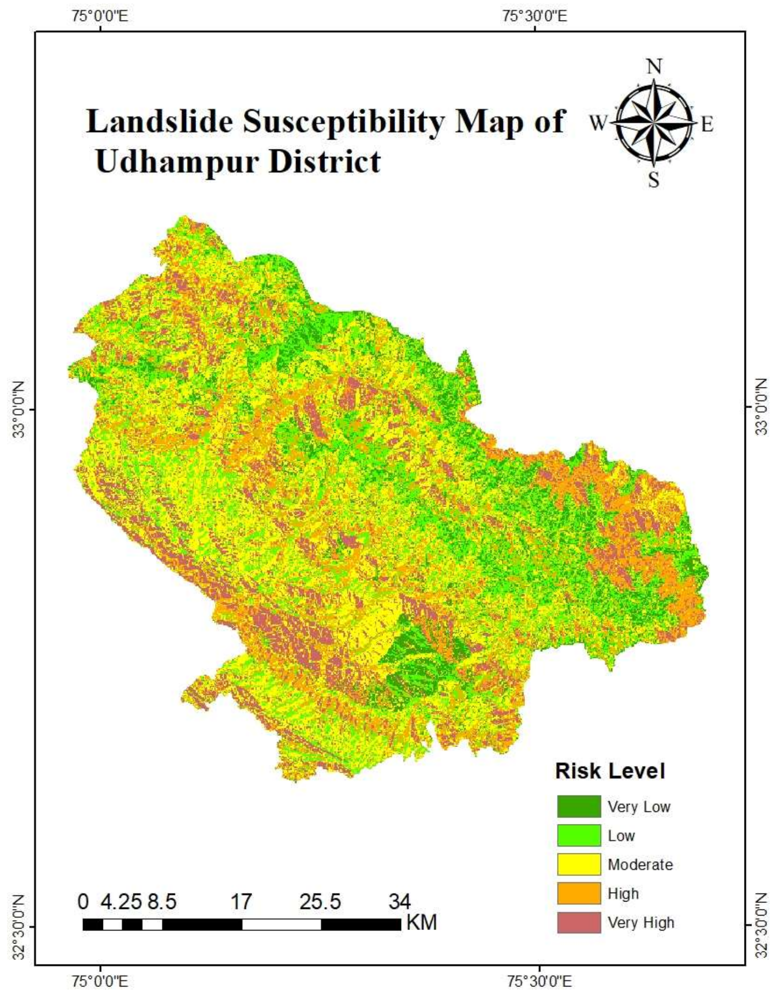



**Figure 15.** Landslide susceptibility map of Udhampur District, classified into five risk categories, highlighting high to very high-risk areas for targeted hazard management and planning.

## 5. Discussion

### 5.1 Accuracy and Interpretation of Landslide Susceptibility

The landslide susceptibility analysis for Udhampur District, derived from the Modified Frequency Ratio (MFR) method, reveals a spatially coherent pattern consistent with the district's geomorphological and hydrological characteristics. As shown in Table 2, decadal rainfall emerges as the dominant factor (FR = 1.41), reflecting the critical role of monsoonal precipitation in triggering slope failures. Elevation (FR = 1.21) and proximity to faults (FR = 1.20) further highlight the combined effects of steep terrain and structural weakness, while slope, NDVI, and LULC show moderate influence, consistent with studies in Himachal Pradesh and Uttarakhand (Bhardwaj et al., 2023; Singh et al., 2022). Distance to roads and streams, though locally significant, exhibit lower FR values, indicating that anthropogenic and fluvial influences are context-dependent rather than uniformly dominant.

The predictive performance, assessed through the ROC curve (Figure 14), yielded an AUC of 0.670, indicating moderate but practically meaningful accuracy. This result is aligned with susceptibility assessments in other complex Himalayan terrains (Sharma et al., 2022; Malekzadeh et al., 2023), where highly heterogeneous topography, spatially clustered landslide occurrences, and limited inventory size constrain maximum predictive accuracy. While the model effectively identifies high-risk zones, areas of medium susceptibility should be interpreted with caution, as local micro-topography or undocumented anthropogenic modifications may influence slope stability.

### 5.2 Spatial Distribution and Risk Patterns

The Landslide Susceptibility Map (**Figure 15**) demonstrates that high to very high-risk zones are predominantly located along steep slopes, near fault lines, and in sparsely vegetated regions experiencing above-average decadal rainfall. Medium-risk zones correspond to moderately steep terrain with mixed land cover, whereas low and very low-risk areas are concentrated in flatter or densely vegetated regions where natural stabilization mechanisms dominate. These spatial patterns underscore the importance of integrating multiple causative factors, including hydrological, topographic, vegetation, and anthropogenic controls, in vulnerability assessment, corroborating findings from the Western Himalaya and other monsoon-affected mountain regions (Dhakal et al., 2020; Bhardwaj et al., 2023).

The analysis highlights that rainfall-induced landslides are the primary concern in Udhampur, consistent with broader Himalayan patterns, yet their interaction with lithology and fault proximity amplifies localized susceptibility. Although lithological heterogeneity was not quantitatively incorporated into the MFR model due to data resolution constraints (**Figure 12**), its contextual interpretation aligns with observed landslide hotspots, particularly in schist- and gneiss-dominated sectors where slope failures are more frequent.

### 5.3 Methodological Implications and Practical Relevance

The integration of Python-based automation with GIS workflows ensures reproducibility, high-resolution mapping, and statistically robust quantification of factor contributions. Normalization, optimal bin-width selection, and continuous FR computation enhance methodological transparency and minimize subjectivity, supporting credible decision-making.

This framework enables targeted hazard management, such as prioritization of high-risk zones for slope stabilization, early warning, or infrastructure planning, and can be adapted for other Himalayan districts with comparable geomorphic and climatic contexts.

The study also demonstrates the value of decadal rainfall integration, which provides a temporally smoothed indicator of hydrological risk. This approach reduces the influence of interannual variability and enhances the reliability of susceptibility estimation, a refinement not always present in prior regional studies.

### 5.4 Limitations and Uncertainty

While the study provides a detailed and reproducible assessment, certain limitations should be acknowledged. The landslide inventory from NASA's Global Landslide Hazard Database represents generalized prone areas rather than all individual events, which may limit fine-scale predictive accuracy. Lithological influence, though contextualized, was not directly incorporated due to spatial resolution constraints. Despite these limitations, the moderate AUC and coherent spatial patterns indicate that the model captures essential factors governing landslide occurrence in Udhampur.

## 6. Conclusion

This study presents the first comprehensive, factor-based landslide susceptibility assessment for Udhampur District, Western Himalayas, integrating a Modified Frequency Ratio (MFR) approach with Python-GIS automation. Key findings include:

1. **Rainfall is the dominant triggering factor**, followed by elevation and proximity to faults, highlighting the combined influence of climatic and structural drivers.
2. **The MFR-based Landslide Susceptibility Index (LSI)** successfully differentiates high-risk zones from low-risk areas, with an AUC of 0.670 confirming moderate predictive performance.
3. **Spatial patterns reveal actionable insights**: steep, sparsely vegetated slopes near faults and high rainfall zones are most susceptible, informing disaster management, infrastructure planning, and targeted mitigation efforts.
4. **Methodological novelty** lies in integrating reproducible Python-based workflows with GIS and decadal rainfall data, enhancing both analytical transparency and practical relevance.

Overall, the study demonstrates that a multi-factor, data-driven susceptibility assessment, grounded in local geomorphology and climate, can support evidence-based hazard mitigation in Himalayan districts. The framework is transferable to similar regions, contributing to more resilient land-use planning, infrastructure design, and community preparedness against recurrent landslide hazards.

**References**


Aleotti, P., & Chowdhury, R. (1999). Landslide hazard assessment: Summary review and new perspectives. *Bulletin of Engineering Geology and the Environment, 58*(1), 21–44. https://doi.org/10.1007/s100640050053

Guzzetti, F., Carrara, A., Cardinali, M., & Reichenbach, P. (1999). Landslide hazard evaluation: A review of current techniques and their application in a multi-scale study, Central Italy. *Geomorphology, 31*(1–4), 181–216. https://doi.org/10.1016/S0169-555X(99)00078-1

Chung, C.-F., & Fabbri, A. G. (2003). Probabilistic prediction models for landslide hazard mapping. *Photogrammetric Engineering & Remote Sensing, 69*(1), 33–41. https://doi.org/10.14358/PERS.69.1.33

Dai, F.-C., Lee, C.-F., & Ngai, Y.-Y. (2002). Landslide risk assessment and management: An overview. *Engineering Geology, 64*(1–2), 65–87. https://doi.org/10.1016/S0013-7952(01)00093-6

Lee, S., & Pradhan, B. (2007). Landslide risk analysis and assessment: A review. *Engineering Geology, 97*(3–4), 137–150. https://doi.org/10.1016/j.enggeo.2008.03.010

Mir, R. A., & Habib, A. (2021). Landslide susceptibility mapping along the NH-44 corridor in Jammu and Kashmir, India. *Environmental Earth Sciences, 80*(6), 1–15. https://doi.org/10.1007/s12665-021-09594-3

Mir, R. A., & Habib, A. (2024). Landslide susceptibility mapping and risk assessment using total estimated susceptibility values along NH44 in Jammu and Kashmir, Western Himalaya. *Natural Hazards, 120*(2), 4257–4277. https://doi.org/10.1007/s11069-024-05672-3

Khan, I., Kainthola, A., & Bahuguna, H. (2024). Landslide susceptibility analysis in the Ramban Basin, Jammu and Kashmir: A statistical approach for comprehensive susceptibility assessment. In *Landslide Risk Assessment and Mitigation in India* (pp. 61–84). Springer. https://doi.org/10.1007/978-981-97-5485-4_6

Sharma, L., & Sharma, P. (2024). Investigating landslide susceptibility in the mountainous area of Jammu and Kashmir using geospatial techniques. *International Journal of Geoscience and Environmental Informatics, 13*(1), 1–15. https://doi.org/10.1504/IJGENVI.2024.141607

Singh, Y., & Sharma, V. (2024). Reservoir induced landslide at Assar, Jammu and Kashmir: A case study. *Journal of the Geological Society of India, 80*(5), 435–439. https://doi.org/10.1007/s12594-024-01798-0

Hussain, G., & Khan, M. A. (2018). Landslide susceptibility mapping along the National Highway-1A in Kashmir Himalaya, India. *Geoscience Frontiers, 9*(5), 1407–1417. https://doi.org/10.1016/j.gsf.2017.12.004

Kumar, A., & Sharma, R. (2017). Assessment of landslide hazards induced by extreme rainfall events in the Jammu region, India. *Geomatics, Natural Hazards and Risk, 8*(3), 241–259. https://doi.org/10.1080/19475705.2017.1303790

Singh, Y., & Sharma, V. (2024). Landslide susceptibility mapping using frequency ratio and Shannon entropy models: A case study from the Chenab Valley, Jammu and Kashmir. *Journal of Earth System Science, 133*(4), 1–15. https://doi.org/10.1007/s12040-024-02367-3

Bose, M., & Sharma, P. (2025). GIS-based multi-criteria analysis for landslide susceptibility mapping in Pulwama district, Jammu & Kashmir, India. *Geoscience Frontiers, 43*(3), 364–374. https://doi.org/10.1016/j.gsf.2025.03.012

Basharat, M., & Khan, M. A. (2023). An assessment of landslide hazards in Muzaffarabad-Azad Jammu & Kashmir using geospatial techniques. *Pakistan Geographical Review, 76*(2), 164–173. https://doi.org/10.1080/19475705.2023.2210255

Kumar, A., & Sharma, R. (2017). Assessment of landslide hazards induced by extreme rainfall events in the Jammu region, India. *Geomatics, Natural Hazards and Risk, 8*(3), 241–259. https://doi.org/10.1080/19475705.2017.1303790

Khan, I., & Kainthola, A. (2024). Landslide susceptibility analysis in the Ramban Basin, Jammu and Kashmir: A statistical approach for comprehensive susceptibility assessment. In *Landslide Risk Assessment and Mitigation in India* (pp. 61–84). Springer. https://doi.org/10.1007/978-981-97-5485-4_6

Pathak, D., & Singh, R. (2016). Knowledge-based landslide susceptibility mapping in the Chamoli-Joshimath area, Garhwal Himalaya, India. *Geoenvironmental Disasters, 3*(1), 1–14. https://doi.org/10.1186/s40677-016-0042-0

Alsabhan, A. H., & Al-Sharhan, M. (2022). Landslide susceptibility assessment in the Himalayan region using geospatial techniques. *Engineering Geology, 292*, 106373. https://doi.org/10.1016/j.enggeo.2021.106373

Kumar, A., & Sharma, R. (2017). Assessment of landslide hazards induced by extreme rainfall events in the Jammu region, India. *Geomatics, Natural Hazards and Risk, 8*(3), 241–259. https://doi.org/10.1080/19475705.2017.1303790

Sharma, L., & Sharma, P. (2024). Investigating landslide susceptibility in the mountainous area of Jammu and Kashmir using geospatial techniques. *International Journal of Geoscience and Environmental Informatics, 13*(1), 1–15. https://doi.org/10.1504/IJGENVI.2024.141607

Singh, Y., & Sharma, V. (2024). Reservoir induced landslide at Assar, Jammu and Kashmir: A case study. *Journal of the Geological Society of India, 80*(5), 435–439. https://doi.org/10.1007/s12594-024-01798-0